

AbhinayaRasa: Digital Classical Dance Education and Academic Management System

A Full-Stack Web Platform for Bharatanatyam Training, Fee Collection, and AI-Powered Decision Support

Bachelor of Science in Information Technology

By

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CHAPTER 1 — INTRODUCTION OF THE PROJECT

1.1 Background

Indian classical dance education involves managing student admissions, curriculum progression, fee payments, and examination materials across multi-year diplomas. Many classical dance academies and fine arts institutions rely on scattered paper forms, offline cash registers, physical brochures, and phone messages, making it difficult for students and parents to view course syllabi, fee deadlines, upcoming events, and past examination papers in a single workspace. A unified digital platform connecting course discovery, student authentication, digital fee receipts, and learning resources is essential for modern fine arts pedagogy [1].

AbhinayaRasa addresses this by combining course discovery, student registration, authenticated dashboards, downloadable syllabus compendiums, examination question paper archives, automated fee receipt generation, and AI-powered inquiry support into one centralized portal (live at <https://abhinaya-qcml.onrender.com>). Students authenticate, review enrolled courses, download syllabus e-books, generate fee receipts, and query the interactive assistant — rather than navigating disconnected channels.

The project is documented as a production-grade academic software system for the **Abhinaya Institute of Classical Bharatanatyam**, under the lineage and guidance of **Guru Dr. Chitra Vishwanathan**. Its digital fee collection and student verification modules maintain full data integrity, keeping the academic focus on how classical cultural education can be preserved, systematized, and delivered to global learners through modern web technology.

1.2 Problem Statement

- Manual and scattered institutional management makes it difficult to track student admissions, verify fee payments, distribute examination question papers, and communicate event schedules. Information exists across physical registers and phone chats, forcing the institution and students to reconcile records manually; a fee payment may be submitted without an instant traceable receipt, or a student may miss an examination structure announcement.
- **AbhinayaRasa** is intended to solve this problem by acting as one authenticated workspace that converts institutional operations into interactive dashboards, downloadable learning compendiums, online fee receipts, and real-time AI assistance. The system makes registration convenient, preserves password encryption and validation, generates uniquely numbered transaction receipts, and communicates academic milestones in a form students can access anytime on desktop or mobile.

1.3 Purpose of the Project

AbhinayaRasa aims to modernize everyday fine arts administration and classical dance learning for students and the academy by connecting institutional records with digital learning resources and automated inquiry support. Rather than requiring users to rely on offline office hours, the system consolidates courses, syllabi, fee structures, receipts, exam archives, and AI assistance inside one responsive cloud platform. This purpose makes AbhinayaRasa a complete information workflow — capture, validation, persistent storage, and digital fulfillment.

Problems with Existing Systems:

- **Scattered records** — admissions, fees, and inquiries are managed across physical notebooks and messaging apps.
- **Manual fee processing** — offline payments lack automated receipt generation, causing verification delays.
- **Unorganized academic resources** — past diploma question papers and syllabus compendiums are not centrally accessible.

- **Fragmented communication tools** — users must contact the office separately for syllabus queries, fees, and workshop dates.
- **Limited 24/7 student guidance** — prospective students seeking admission criteria outside business hours face delayed replies.
- **Absence of student portal** — lack of authenticated personal student dashboards to access enrolled course tiers.

How the Proposed System Solves These Problems

AbhinayaRasa addresses each of these gaps within a single authenticated web ecosystem. Course tiers (Foundation, Diploma, Nritya Prabha, Master Diploma, Visharad) are structured with transparent curriculum and fee brackets. Student accounts persist securely in a MySQL database with bcrypt password hashing. When fee payments are submitted, the system automatically computes dues, issues a unique transaction ID (TXN...), and generates a downloadable digital HTML/PDF receipt. The 121 MB *Bharatanatyam Learning Compendium* and past question papers (2017–2024) are organized into instant digital downloads, while an integrated AI Chat Assistant resolves student inquiries instantly without human delay.

1.4 Objectives of the Project

The following objectives guide the development of AbhinayaRasa. Each objective is specific, measurable through the corresponding module, achievable within the current implementation, relevant to the identified problems, and scoped to the project timeline.

- To provide secure authenticated access to the system through credential validation, bcrypt hashing, and session management.
- To enable students to explore classical Bharatanatyam courses, academic calendars, and curriculum requirements.
- To automate fee payment processing with dynamic fee calculations and downloadable digital receipt generation.
- To deliver centralized digital learning archives, including the complete Bharatanatyam compendium and past question papers.
- To deploy an interactive 24/7 AI Dance Chatbot Assistant for instant admission, syllabus, and administrative guidance.
- To establish a secure administrative management interface for reviewing live student records, transactions, and inquiries.

1.5 Scope of the Project

In-Scope

The following table lists the verified in-scope AbhinayaRasa features, presented in the logical order in which a user encounters them.

Feature	Verified Function
Authentication & Profile	Student registration, login, logout, password recovery, session validation, and enrolled course tracking.
Home & Heritage Showcase	Hero banner with classical dance imagery, institute legacy, guru lineage, and quick portal links.
Courses & Programs	Detailed syllabus breakdown for Foundation, Diploma, Nritya Prabha, Master Diploma, and Visharad.
Student Portal & Dashboard	Authenticated dashboard providing quick navigation to enrolled course resources and announcements.

Feature	Verified Function
Fee Payment & Receipts	Course-wise fee selection, transaction ID generation, and automatic downloadable HTML/PDF receipt generation.
Academic E-Book Compendium	Direct digital access and download of the complete 121 MB <i>Bharatanatyam Learning Compendium</i> .
Examinations Hub	Previous years' examination question papers (2017–2024), mark sheets, and theoretical patterns.
AI Dance Assistant Chatbot	Interactive AI conversational widget answering queries regarding admissions, fees, syllabus, and schedules.
Contact & Inquiries	Interactive inquiry submission persisting message records to MySQL with instant user feedback.
Admin DB Manager	Passcode-protected dashboard allowing administrators to inspect live users, payments, and message counts.

Future Scope

The items below extend the current implementation in a logical progression, building on structures that already exist in the system:

- **Live Payment Gateway Integration** — direct Razorpay / Stripe webhook processing beyond generated receipts.
- **AI Posture Analysis** — computer vision pose estimation for student *Adavus* and *Hastas* practice.
- **Live Virtual Classrooms** — WebRTC / Zoom live video streaming integration for remote masterclasses.
- **SMS / WhatsApp Notifications** — automated transaction and exam reminder broadcasts.

Out-of-Scope / Exclusions

The following are intentionally excluded from the current implementation:

- Commercial banking advisory or financial credit management.
- Native iOS and Android application store binaries.
- Automated biometric attendance hardware integration.

CHAPTER 2 — LITERATURE SURVEY

Academic and cultural web applications commonly combine informational showcases with student authentication, resource downloads, and automated administration. Two established platforms were reviewed to identify concepts relevant to AbhinayaRasa.

2.1 YG Entertainment Official Website

Brief Description

The YG Entertainment Official Website is the digital platform of one of South Korea's leading cultural and entertainment agencies. Its official feature structure includes high-definition artist portfolios, dynamic media banners, multilingual navigation, event schedules, and discography archives [2]. Its design language emphasizes immersive visual storytelling and modern dark-mode aesthetics.

Features List

- High-resolution multimedia showcase with smooth transition animations.
- Interactive navigation bar with structured dropdown menus.
- Responsive layout engineered for seamless desktop and mobile rendering.
- Modern typographic hierarchy paired with premium dark and gold color accents.

Technology Overview

The table below reflects the general technology categories that a cross-platform multimedia showcase product relies on.

Technology Category	Typical Role
Web Client Applications	Deliver responsive, high-fidelity media screens across desktop and mobile devices.
Frontend Styling & Animations	CSS3 animations, glassmorphism, and responsive grid layouts.
Content Delivery Network	Serves global multimedia assets with low latency.
Interactive Scripts	Provides client-side carousels, video player modals, and smooth scroll effects.

Observation

YG Entertainment demonstrates that digital arts portals require rich visual elegance and immediate user engagement. AbhinayaRasa applies this principle by utilizing a classical Temple Gold (#d4af37) and Royal Maroon (#8b0000) palette, Cinzel typography, and micro-interactions. AbhinayaRasa broadens the fine-arts design by adding full-stack academic capabilities — student authentication, examination paper downloads, fee receipts, and an AI dance assistant.

2.2 Sathaye College Official Website

Brief Description

The Sathaye College Official Website serves as the central academic information portal for higher education students, faculty, and parents. It organizes syllabus frameworks, examination notices, fee structures, departmental circulars, and institutional contact directories across multi-faculty programs.

Features List

- Department-wise curriculum, course regulations, and syllabus outlines.
- Examination circulars, past question paper archives, and timetable downloads.
- Online enquiry forms, admission guidelines, and administrative notices.
- Structured academic calendar tracking yearly milestones and festivals.

Technology Overview

The table below lists the general technology categories an institutional academic portal typically depends on.

Technology Category	Typical Role
Academic Web Layouts	Presents syllabus, notices, and examination archives in a structured format.
Database Management	Maintains student admissions, course records, and departmental contacts.
Document Repositories	Distributes official circulars, syllabi, and question papers via PDF downloads.
Web Form Processing	Captures student enquiries and feedback submissions.

Observation

Sathaye College illustrates how educational institutions must organize academic services clearly for ease of student navigation. AbhinayaRasa adapts these academic practices for classical Bharatanatyam education by structuring diploma question papers, marksheet frameworks, and the *Bharatanatyam Learning Compendium* into intuitive digital categories.

CHAPTER 3 — SYSTEM REQUIREMENTS AND DESIGNING

3.1 System Modules

AbhinayaRasa is divided into integrated modules. Frontend pages represent user tasks, PHP backend scripts process domain logic, and MySQL tables persist user records. The Registration and Login modules form the secure authentication boundary.

3.1.1 Registration Module

The Registration Module allows a new student to create an authenticated AbhinayaRasa account. It works step by step as follows:

Step 1 — Display Registration Page: the user is shown a registration form requesting full name, email, course level, and password.

Step 2 — Enter Registration Details: the user fills in the required fields and agrees to the terms.

Step 3 — Submit Registration Request: the form data is sent via POST to register.php, passing through input sanitation and regex field validation.

Step 4 — Validate Input: the system confirms the email is not already registered in the users table; if it is, an *'Email already registered'* error is returned.

Step 5 — Hash Password: the password is encrypted using PHP `password_hash(..., PASSWORD_DEFAULT)` (bcrypt); plain text is never persisted.

Step 6 — Create User Record: the new student record is inserted into the MySQL users table via PDO prepared statements.

Step 7 — Issue Flash Message and Redirect: a session success message is set and the student is redirected to login.php to authenticate.

3.1.2 Login Module

The Login Module authenticates an existing student using registered email and password. It works step by step as follows:

Step 1 — Display Login Page: the user is shown the login interface at login.php.

Step 2 — Enter Login Credentials: the student inputs their registered email address and password.

Step 3 — Submit Login Request: form data is submitted via POST to login.php, passing through validation.

Step 4 — Validate Existence: the system queries the users table by email; if no matching user exists, an *'Invalid email or password'* response is returned.

Step 5 — Verify Password: the submitted password is verified against the stored bcrypt hash using `password_verify()`; if invalid, the attempt is rejected.

Step 6 — Initialize Session: on success, \$_SESSION['user_id'], full_name, and email are set in server memory.

Step 7 — Redirect to Dashboard: the student is redirected to dashboard.php, granting access to academic resources, exam archives, and fee services.

Module Summary

Module	Purpose and Principal Functions
Authentication & Profile	Registration, login, logout, password recovery, session verification, and course tracking.
Academic Portal & Dashboard	Authenticated dashboard displaying enrolled course details, announcements, and direct action cards.
Online Fee & Receipts	Course-wise fee selection, transaction ID generation, and automatic downloadable HTML/PDF receipt creation.
Examinations & E-Book	121 MB Bharatanatyam Learning Compendium download, past question papers (2017–2024), and mark sheets.
AI Dance Chatbot Assistant	Interactive 24/7 AI chat widget answering natural-language queries regarding syllabus, fees, and guru.
Admin DB Manager	Passcode-protected manager inspecting real-time student counts, payment receipts, and contact messages.

Beyond authentication, AbhinayaRasa's modules fall into three groups:

1. **The core academic modules** — course directories, syllabus outlines, and e-book compendiums — provide foundational learning materials.
2. **The transaction and evaluation modules** — fee payment, unique receipt generation, and examination paper archives — automate administration.
3. **The engagement and assistance modules** — AI Chatbot, contact inquiry desk, and admin database manager — ensure continuous support.

3.2 Hardware and Software Requirements

The system requires a standard development workstation, PHP 8.x runtime, MySQL database service, and modern web browsers.

3.2.1 Hardware Requirements

Component / Area	Verified Requirement	Purpose
Processor	Dual-core 64-bit processor, 2.0 GHz or higher	Runs Apache web server, PHP engine, and MySQL service.
RAM	4 GB minimum; 8 GB recommended	Supports concurrent local development and database tasks.
Storage	At least 5 GB free space	Stores application source, 121 MB PDF compendium, and receipts.
Display	1366 × 768 minimum; 1920 × 1080 recommended	Ensures clear viewing of responsive layouts and dashboards.
Internet Connection	Stable broadband / 4G / 5G	Supports cloud deployment on Render and remote database queries.
Mobile Device Testing	Recent Android / iOS device or browser emulator	Verifies responsive mobile navigation and chat widget.

3.2.2 Software Requirements

Area	Verified Requirement
Operating System	Windows 10/11, macOS, or Linux capable of running Apache, PHP, and MySQL.
Frontend Technologies	HTML5 semantic elements, CSS3 (Glassmorphism, Gold & Maroon styling), and Vanilla JavaScript.
Backend Framework & Language	PHP 8.x with PDO and MySQLi extensions, session handling, and modular controllers.
Database Service	MySQL 8.0 / MariaDB relational database (abhinaya_db).
Runtime Environment	Apache 2.4 web server (XAMPP for local testing / Docker container on Render Cloud).
Web Browsers	Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari with JavaScript and cookies enabled.
Development Tools	Antigravity IDE, VS Code, Git version control, and phpMyAdmin database manager.

3.2.3 Technology Stack Summary

- **Frontend:** HTML5, CSS3, JavaScript (ES6+), FontAwesome 6.5, Google Fonts (Cinzel & Poppins).
- **Backend:** PHP 8.x with prepared statements and session-based route security.
- **Database:** MySQL relational database managing users, payments, and contacts tables.
- **Cloud Deployment:** Render Cloud platform hosting containerized Apache/PHP at <https://abhinaya-qcml.onrender.com>.

3.3 Planning and Scheduling

AbhinayaRasa was developed progressively from June 2026 to August 2026. Planned and actual work is shown below; blue indicates the planned schedule, green indicates actual completion, and amber marks variance.

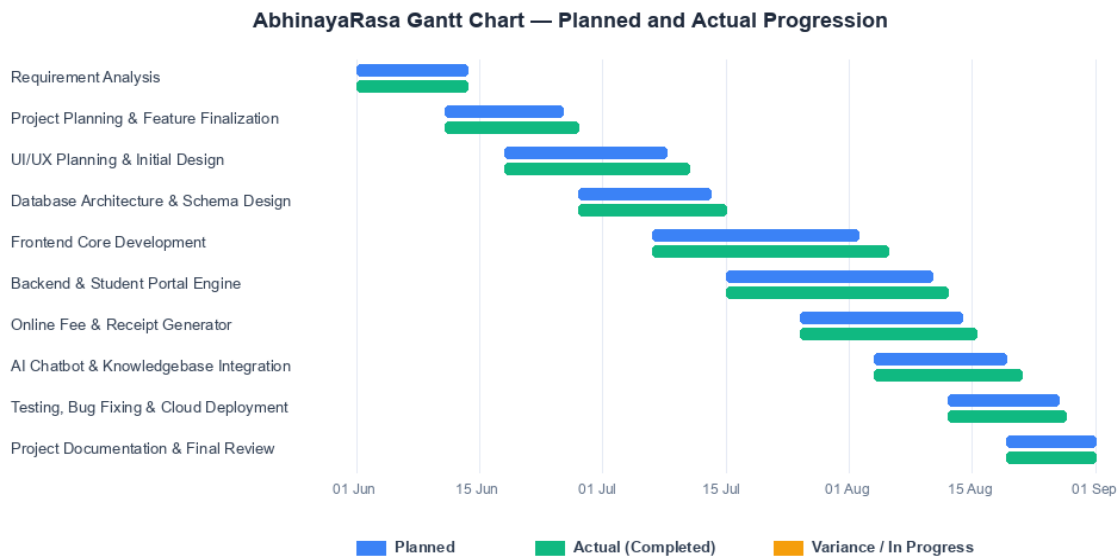


Figure 3.1: Gantt Chart of AbhinayaRasa Development

3.3.1 Gantt Chart — Planned vs Actual (Exact Calendar Dates)

All dates are preserved exactly from the verified project schedule.

Activity	Planned Start	Planned End	Actual Completion
Requirement Analysis	10 Jun 2026	15 Jun 2026	15 Jun 2026
Project Planning & Feature Finalization	16 Jun 2026	22 Jun 2026	23 Jun 2026
UI/UX Planning & Initial Design	20 Jun 2026	30 Jun 2026	02 Jul 2026
Database Architecture Design	24 Jun 2026	02 Jul 2026	03 Jul 2026
Frontend Core Development	01 Jul 2026	15 Jul 2026	17 Jul 2026
Backend & Student Portal Engine	05 Jul 2026	20 Jul 2026	22 Jul 2026

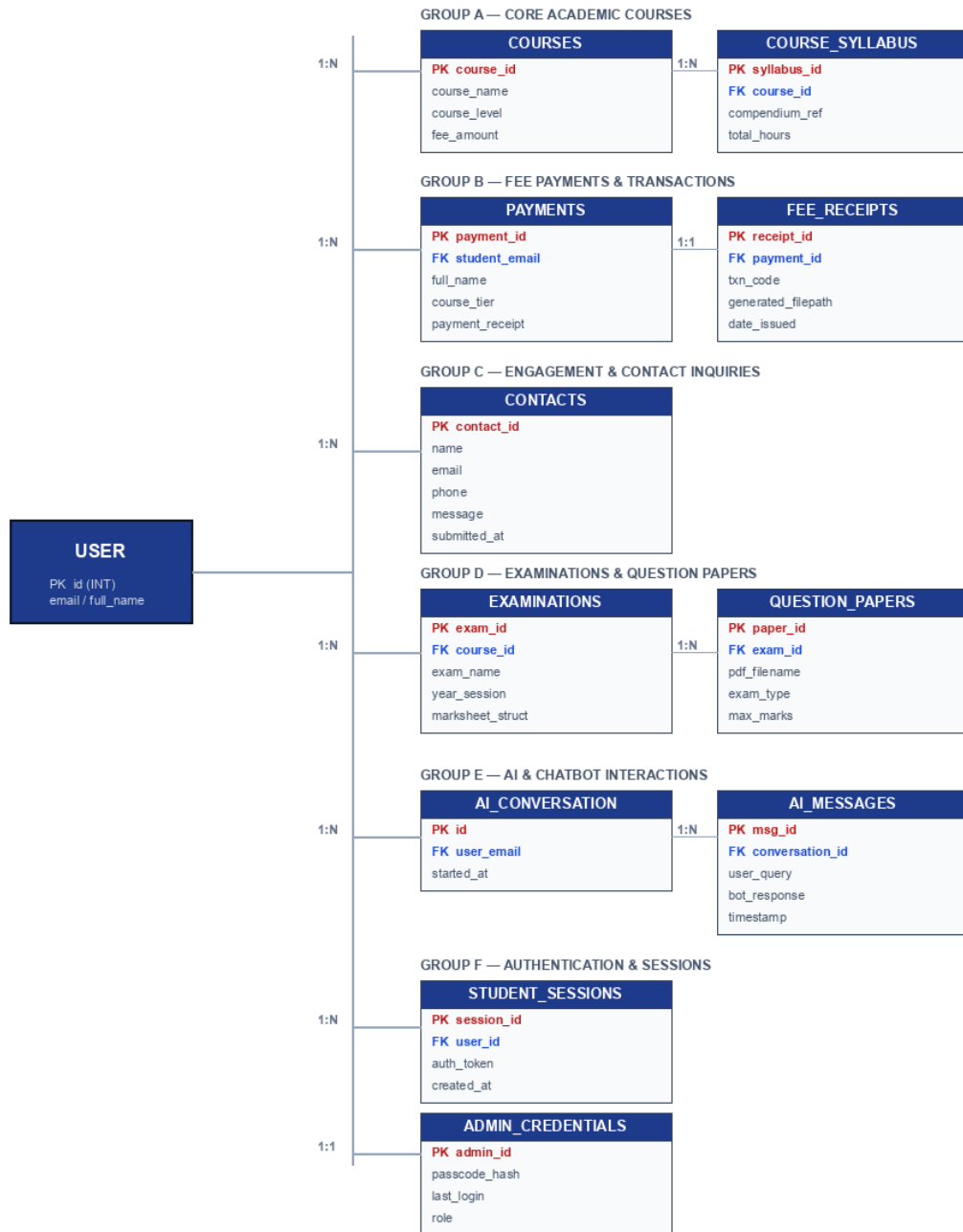
Activity	Planned Start	Planned End	Actual Completion
Online Fee & Receipt Generator	18 Jul 2026	27 Jul 2026	28 Jul 2026
Examinations Hub & E-Book Integration	22 Jul 2026	31 Jul 2026	02 Aug 2026
AI Dance Chatbot Assistant	27 Jul 2026	06 Aug 2026	07 Aug 2026
Admin DB Inspection Module	01 Aug 2026	12 Aug 2026	14 Aug 2026
Mobile Responsiveness & UI Polish	07 Aug 2026	16 Aug 2026	18 Aug 2026
Testing, Bug Fixing & Cloud Deployment	20 Aug 2026	31 Aug 2026	31 Aug 2026
Project Documentation & Final Review	28 Aug 2026	05 Sep 2026	Completed

3.4 Conceptual Models

Conceptual Models are high-level visual representations of a system's structure, data, and behavior, without code-level detail. In this project, they include the ER Diagram, Data Dictionary, Data Flow Diagrams, Use Case Diagram, and Activity Diagram, showing how data is organized, how it flows through the system, and how key processes like login, course enrollment, and fee payments behave.

3.4.1 ER Diagram

Central USER ownership and the main stored-data relationships are shown with readable PK/FK summaries. USER is the primary ownership entity; related entities are organized into logical groups (core academic courses, fee payments & receipts, engagement & inquiries, examinations, AI chatbot interactions, and authentication sessions) so that ownership and cardinality remain clear without overloading the diagram with every database attribute.



Note: Calculated views (Dashboard, Syllabus Calendar, Fee Calculator) are derived views and not independent database tables.

Figure 3.2: Entity Relationship Diagram of AbhinayaRasa

Note: Calculated views such as Dashboard, Syllabus Calendar, and Fee Calculator are derived views and are not independent database tables.

3.4.2 Data Dictionary

The verified schema contains normalized tables for the AbhinayaRasa portal; each row below summarizes a table's key fields and purpose rather than listing every column, for readability.

Table	Key Fields	Purpose
users	id (PK), full_name, email, password, course, created_at	Student credentials, encrypted password hash, enrolled course, and registration date.
courses	course_id (PK), course_name, course_level, duration, fee_amount	Dance program tier definitions, syllabus level, and associated tuition amounts.
payments	id (PK), full_name, email (FK), course, payment_receipt, created_at	Fee payment records, student link, course tier, and generated receipt file path.
fee_receipts	receipt_id (PK), payment_id (FK), txn_code, generated_filepath, date_issued	Generated receipt metadata with unique transaction reference numbers.
contacts	id (PK), name, email, phone, message, submitted_at	User inquiry messages, student questions, and submission timestamps.
examinations	exam_id (PK), course_id (FK), exam_name, year_session, marksheet_struct	Examination patterns, evaluation criteria, and marksheet guidelines.
question_papers	paper_id (PK), exam_id (FK), pdf_filename, exam_type, max_marks	Downloadable past examination question paper PDF references (2017–2024).
student_ebooks	ebook_id (PK), title, pdf_path, filesize, access_level	121 MB Bharatanatyam Learning Compendium repository references.

Table	Key Fields	Purpose
ai_conversations	id (PK), user_email (FK), started_at	Groups student AI chat questions into active conversation sessions.
ai_messages	msg_id (PK), conversation_id (FK), user_query, bot_response, timestamp	Individual chat queries and dynamic rule-based answers generated by the chatbot.
admin_credentials	admin_id (PK), passcode_hash, last_login, role	Master admin passcode authentication for the database inspection portal.

3.4.3 Data Flow Diagram — Level 0

The context diagram places **ABHINAYARASA WEB PORTAL & ACADEMIC SYSTEM** as the single central process (0.0), with the user on the left and external entities (Institute Admin/Guru, AI Chatbot/Repository, and MySQL Database Storage) arranged to the right so that flows read left to right without crossing lines.

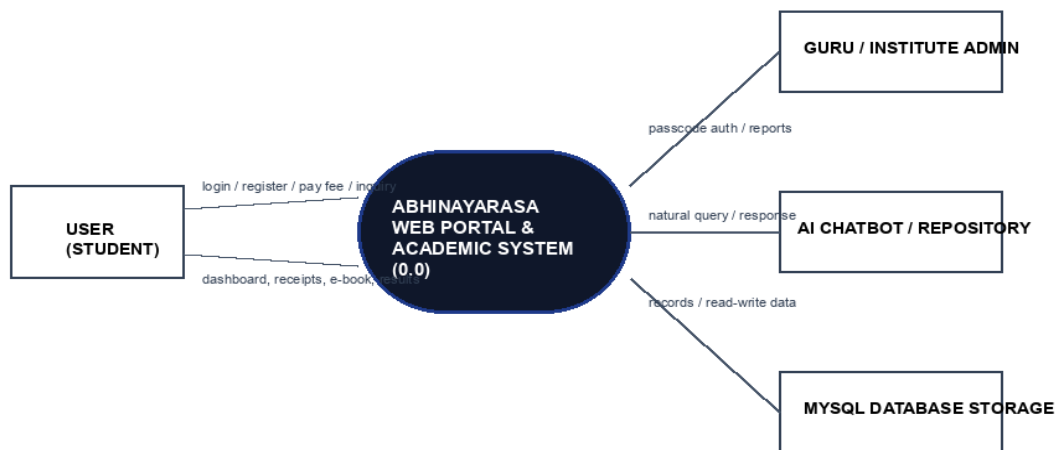


Figure 3.3: Data Flow Diagram — Level 0

3.4.4 Data Flow Diagram — Level 1

Seven numbered processes connect to four data stores. The user column sits on the left with direct flows into each process; processes read and write to specialized data stores and return computed dashboards, receipts, and e-books.

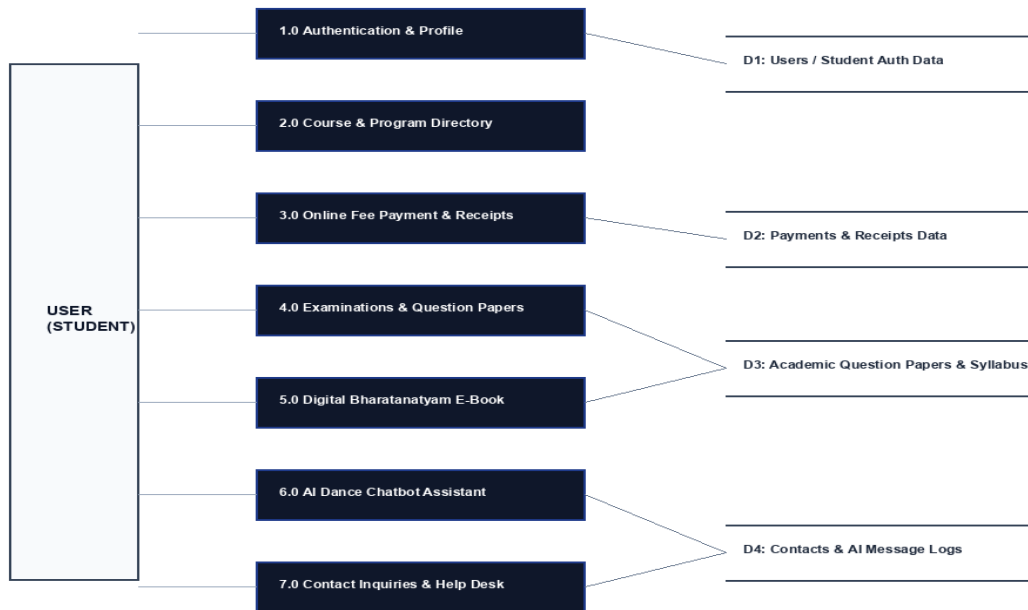


Figure 3.4: Data Flow Diagram — Level 1

Explanation: Process 1.0 authenticates students against D1 (Users Auth Data). Process 3.0 computes fee dues and writes to D2 (Payments Data). Processes 4.0 and 5.0 retrieve PDF files from D3 (Academic Store), while Processes 6.0 and 7.0 manage inquiries and logs in D4.

3.4.5 Use Case Diagram

The primary USER actor interacts with AbhinayaRasa through use cases grouped by function: account, academic courses, fee & receipts, exams & e-book, AI & engagement, and admin management. Grouping keeps related use cases together and avoids overlapping lines.



Figure 3.5: Use Case Diagram of AbhinayaRasa

3.4.6 Activity Diagram

The activity diagram follows the complete user workflow from authentication through module action, result review, continuation, and logout, arranged left to right in standard UML flow-chart notation.

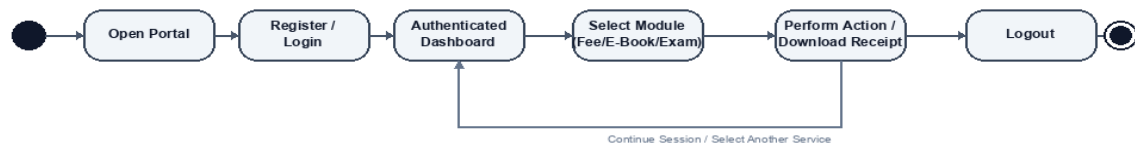


Figure 3.6: Activity Diagram of AbhinayaRasa

Perform Action may represent course enrollment, fee payment & receipt download, compendium reading, past question paper download, or chatbot query.

CONCLUSION

AbhinayaRasa demonstrates how a fine arts educational platform can move beyond simple static brochures to become a connected, interactive academic workspace for classical Bharatanatyam students and teachers. By bringing course curricula, authenticated student dashboards, downloadable syllabus compendiums, examination question papers, automated digital fee receipts, and an intelligent AI Dance Chatbot into a single authenticated platform, the project addresses the core operational problems identified at the outset — that scattered records and offline processes make it difficult to access dance education efficiently.

The system was built using a modern, maintainable technology stack — PHP 8.x with PDO prepared statements, MySQL relational persistence, and a responsive frontend styled with classical Gold and Maroon aesthetics. The literature survey of YG Entertainment and Sathaye College helped position AbhinayaRasa's design choices, while the project's own database schema, module structure, and diagrams (Gantt, ER, DFD Level 0, DFD Level 1, Use Case, and Activity) reflect an implementation that was verified against the live cloud-deployed system.

Overall, the project meets its stated objectives: it replaces manual paperwork with a centralized digital ecosystem, supports structured learning, and demonstrates a complete information workflow — from data capture to fulfillment — suitable for a B.Sc. IT academic field project.

Note:

The deployed cloud platform was verified live at <https://abhinaya-qcml.onrender.com/>. Its public landing pages, authentication boundary, examination repositories, and fee receipt generation executed with verified database persistence.